

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

23,262

Developers surveyed

8 Years

of longitudinal data

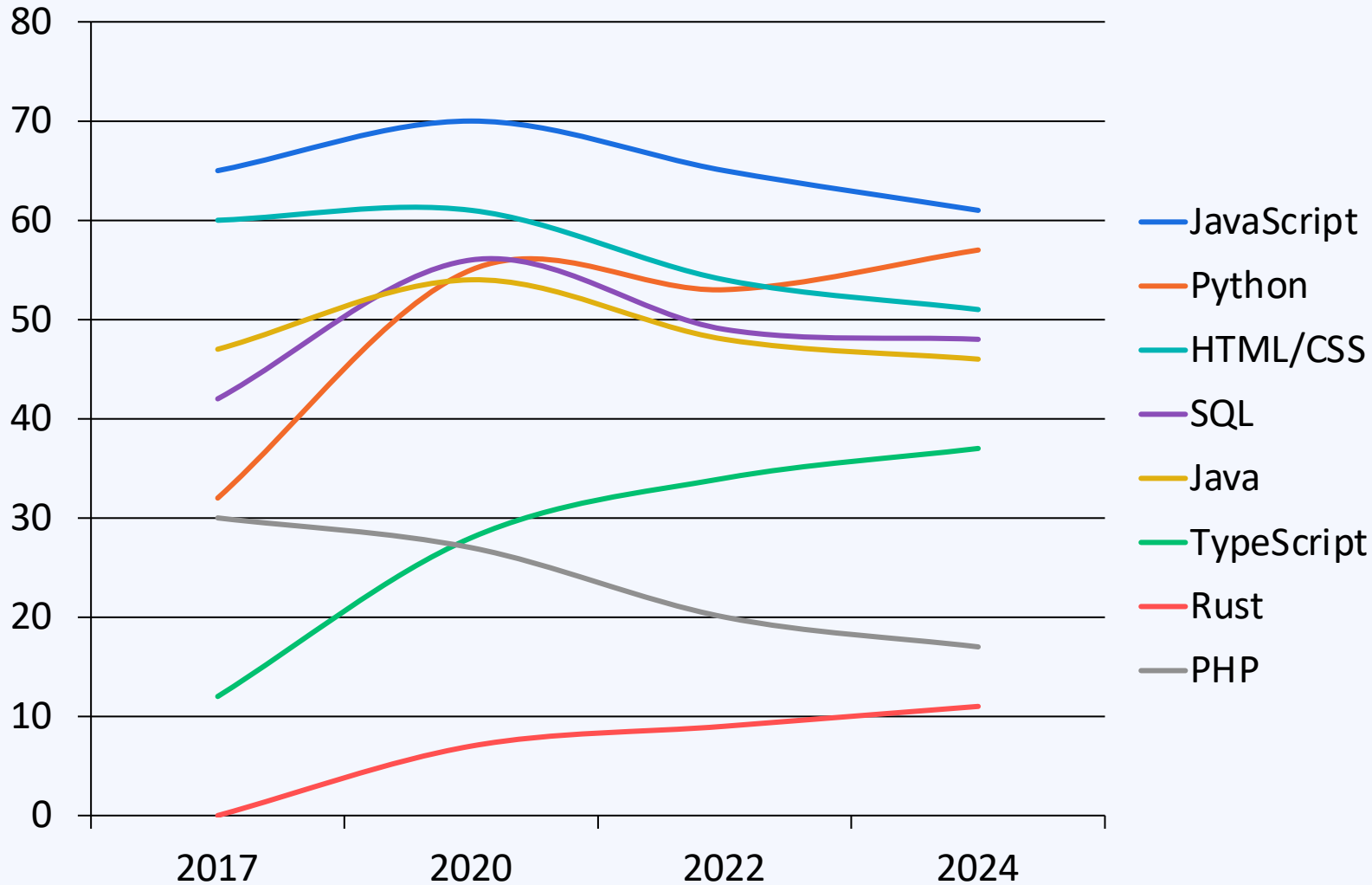
20+

Languages tracked

JetBrains Annual Survey · 2024 Edition

Prepared for Engineering Leaders, CTOs & Developer Experience Teams

Python Closes the Gap: 8 Years of Language Usage Shifts



 Python

32% → 57%

 JavaScript

65% → 61%

 TypeScript

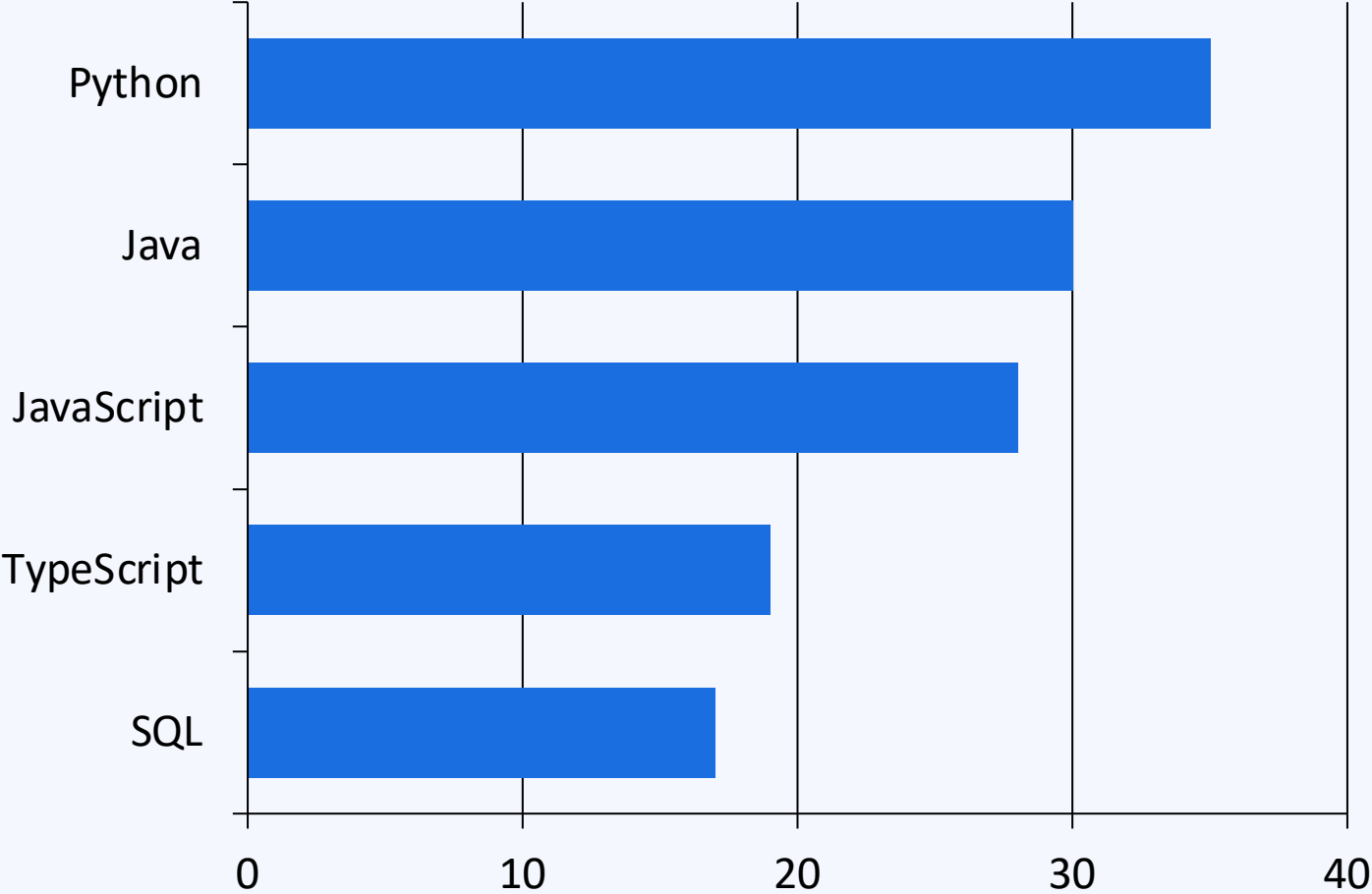
12% → 37%

 PHP

30% → 17%

Python Is Now the #1 Primary Language at 35%

Primary Language (%)

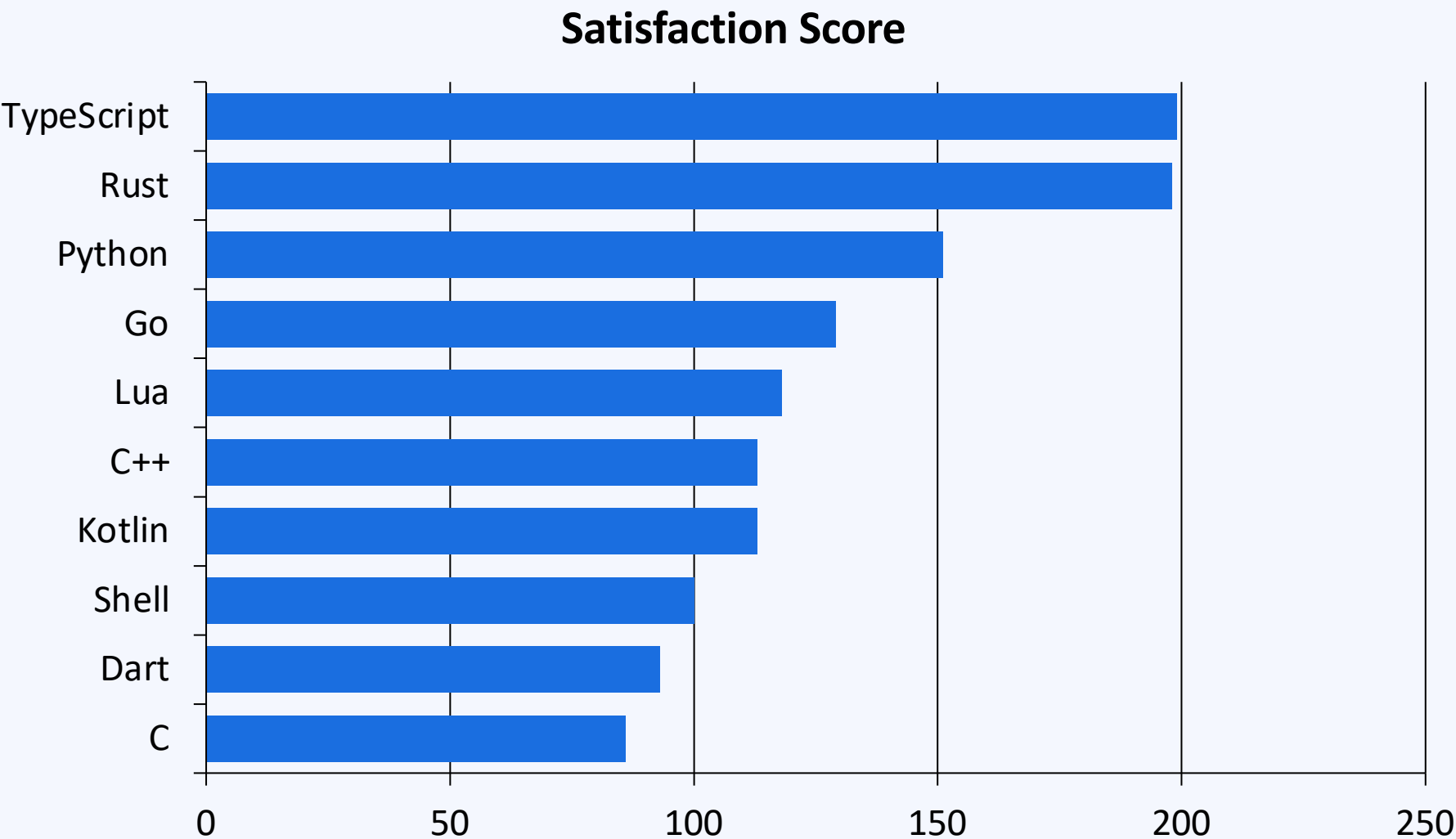


Adoption Intent (plan to switch primary)

Python	6%
TypeScript	5%
Java	4%
SQL	4%
Kotlin	3%

Adoption intent stays below 6% across all languages — ecosystem lock-in is real.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Top Performers

🥇 TypeScript

199

Type-safe, modern DX

🥈 Rust

198

Memory-safe, blazing fast

🥉 Python

151

AI/ML powerhouse

4th Go

129

Cloud-native darling

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Top Language	Share	Insight
AI / Machine Learning	Python	31%	Undisputed AI/ML leader
Data Science & Analytics	Python	28%	Growing with AI boom
Data Processing	Python	17%	ETL & pipeline default
Integrating with APIs	Python	38%	Glue code & automation
User Interface	TypeScript	62%	React/Vue/Angular stack
APIs & Services	Java	49%	Spring Boot dominance
Application Logic & Workflows	Java	55%	Enterprise backend
System Tools & Components	C++	27%	Performance-critical code

What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML Teams

Python leads AI (31%), data science (28%), and data processing — no close second. Prioritise Python expertise in data-focused hires.

02

TypeScript Is the Future of Frontend & Full-Stack

TypeScript owns 62% of UI work and satisfaction ranks #1 at 199. Standardise on TypeScript for web-facing teams.

03

Rust & Go Are the Infrastructure Growth Plays

Both designed for performance and concurrency. Rust climbing to 11%, Go stable at 18%. Strong adoption intent signals long-term momentum.

04

PHP & Ruby Hiring Pools Are Shrinking

PHP: 30%→17%; Ruby: 10%→4% over 8 years. Plan migrations and account for reduced candidate pipelines.

05

Type Safety Correlates with Developer Retention

TypeScript (199) and Rust (198) top satisfaction charts. Prioritising type-safe languages is a retention strategy, not just a quality decision.